MAT1375, Classwork14, Fall2026

Ch13. Exponential and Logarithmic Functions II

ln : $\log_e$

1. Rewrite the equation in its equivalent exponential form.

a) $x = \log_2(16)$. b) $2 = \log_5 x$. c) $x = \log_{13}(1)$. d) $x = \ln(e^7) = \log_e(e^7)$

a) $2^x = 16 \Rightarrow x = 4$

b) $5^2 = x \Rightarrow x = 25$

c) $13^x = 1 \Rightarrow x = 0$

d) $e^x = e^7 \Rightarrow x = 7$

2. Evaluate the expression by rewriting it as an exponential expression.

a) $\log_5(125)$. b) $\log_4(1)$. c) $\log_7\left(\frac{1}{49}\right)$. d) $\log_2(\sqrt[5]{2})$. e) $\log_{25}(5)$.

a) Let $x = \log_5(125)$, $5^x = 125$, $x = 3$

b) $y = \log_4(1)$, $4^y = 1$, $y = 0$

c) $x = \log_7\left(\frac{1}{49}\right)$, $7^x = \frac{1}{49} = \frac{1}{7^2} = 7^{-2}$, $x = -2$

d) $x = \log_2(\sqrt[5]{2})$, $2^x = \sqrt[5]{2} = 2^{\frac{1}{5}}$, $x = \frac{1}{5}$

e) $x = \log_{25}(5)$, $25^x = 5 = (25)^{\frac{1}{2}}$, $x = \frac{1}{2}$

3. Graph $f(x) = 3^x$ and $g(x) = \log_3 x$ in the same coordinate.

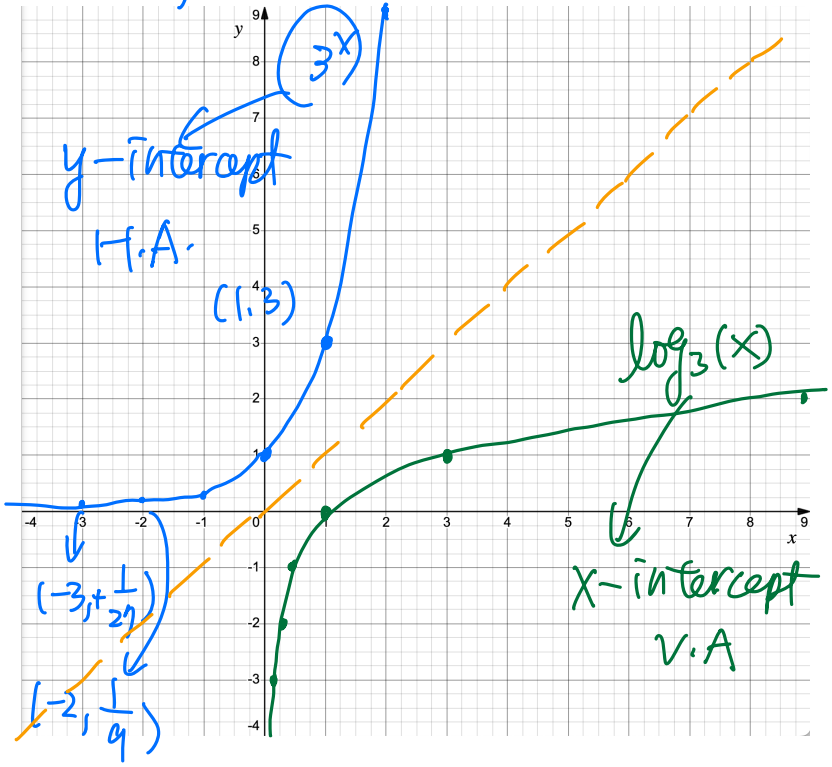
$y = 3^x \Rightarrow x = 3^y \Rightarrow \log_3(x) = y \Rightarrow f^{-1}(x) = \log_3(x)$

$(-\infty, \infty)$, $(0, \infty)$

x	-3	-2	-1	0	1	2
$f(x)$	3^{-3}	3^{-2}	3^{-1}	3^0	3^1	3^2
	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9

$(0, \infty)$, $(-\infty, \infty)$

x	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9
$g(x)$	-3	-2	-1	0	1	2



$-3 = \log_3 x \Rightarrow x = 3^{-3} = \frac{1}{3^3} = \frac{1}{27}$

$-2 = \log_3 x \Rightarrow x = 3^{-2} = \frac{1}{9}$

$-1 = \log_3 x \Rightarrow x = 3^{-1} = \frac{1}{3}$

$0 = \log_3 x \Rightarrow x = 3^0 = 1$

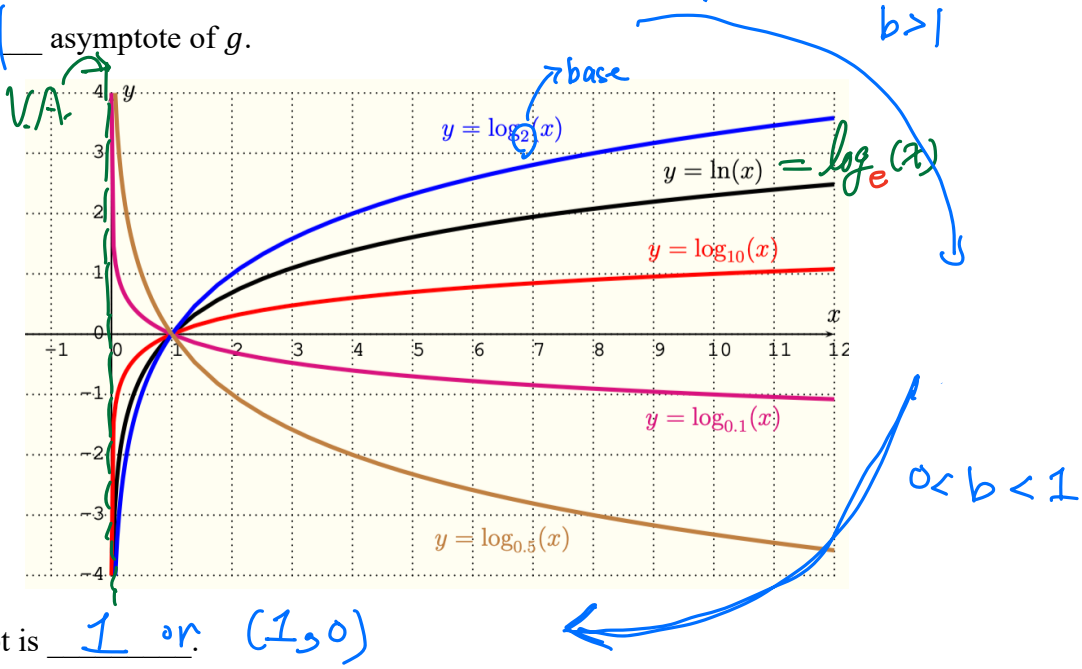
$1 = \log_3 x \Rightarrow x = 3^1 = 3$

$2 = \log_3 x \Rightarrow x = 3^2 = 9$

4. Characteristics of ~~Exponential~~ log Function of $g(x) = \log_b x$.

(a) The domain of g: $(0, \infty)$; The range of g: $(-\infty, \infty)$ or All Real numbers

(b) There is NO y-intercept. In fact, g approaches, but never touches y-axis which is a vertical asymptote of g.



(c) Its x-intercept is 1 or $(1, 0)$.

(d) For $b > 1$, $g(x) \to \infty$ as $x \to \infty$, $g(x) \to -\infty$ as $x \to 0^+$.

(e) For $0 < b < 1$, $g(x) \to -\infty$ as $x \to \infty$, $g(x) \to \infty$ as $x \to 0^+$.

5. **Basic logarithmic evaluations**: Let $f(x) = b^x$ and $g(x) = \log_b x, b > 0, b \neq 1$. We have

(1) Elementary logarithms: $b = b^1 \Leftrightarrow 1 = \log_b(b)$. ($\log_b b$)

$1 = b^0 \Leftrightarrow 0 = \log_b(1)$. ($\log_b 1 = 0$)

(2) Inverse properties: $b^{\log_b(x)} = x$. $(f(g(x)) = x)$ $2^{\log_2(3)} = 3$

$\log_b(b^x) = x$. $(g(f(x)) = x)$ $\log_5(5^3) = 3$

(3) Change-of-Base property: **10-base**: $\log_b(x) = \frac{\log_{10}(x)}{\log_{10}(b)} = \frac{\log(x)}{\log(b)}$.

$\log_{10}(5)$ or $\log_{10}(100)$, $\log(5)$, $\log(100)$; $\ln(100) = \log_e(100)$

natural base: $\log_b(x) = \frac{\log_e(x)}{\log_e(b)} = \frac{\ln(x)}{\ln(b)}$.

6. Given $f_1(x) = \log_e(x), f_2(x) = \log_{0.5}(x), f_3(x) = \log_{10}(x), f_4(x) = \log_2(x)$, and $f_5(x) = \log_{0.1}(x)$. Using the following numbers to find the order of these five functions from small to larger for a fixed $x > 1$:

$\ln(2) = 0.6931, \ln(10) = 2.3026, \ln(0.1) = -2.3026, \ln(0.5) = -0.6931$.

$f_2(x) = \log_{\frac{1}{2}}(x) = \frac{\ln(x)}{\ln(\frac{1}{2})} = \frac{\ln(x)}{-0.6931}$ $f_5(x) = \log_{0.1}(x) = \frac{\ln x}{-2.3026}$

$f_1(x) = \ln(x)$

$f_3(x) = \log_{10}(x) = \frac{\ln(x)}{\ln(10)} = \frac{\ln(x)}{2.3026}$, $f_4(x) = \log_2(x) = \frac{\ln x}{\ln 2} = \frac{\ln(x)}{0.6931}$

$$\frac{\ln(x)}{-0,6931} < \frac{\ln(x)}{-2,3026} < \frac{\ln(x)}{2,3026} < \ln x < \frac{\ln x}{0,6931}$$

$$f_2 < f_5 < f_3 < f_1 < f_4$$